Rank-27 Active-Decision Representation Repair

Tactix Development Project

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Executive summary

DefaultPolicy already enters ordinary search and warming decisions only while `unassigned_active` is positive. Current Kissat relies on the same invariant in its focused queue and stable heap selectors. Tactix instead sent those calls through the policy-neutral N1 selector, carrying impossible end-of-structure fallbacks through every ranked scan. The repair gives DefaultPolicy active-invariant queue/heap entry points while retaining the total generic selector for public and injected policies.

The frozen isolated prototype diff has SHA-256 `5f64e78a5878a8b27a7097e9e40aeda853c4f42941b405e0448e4f731387a8c0`; the commit-relative source diff over pushed BVE base `a07bc93` has SHA-256 `a5662046a26f7ec3acfa802d805893d136f49c2af4317457bc3be3ea17c54ad0`; the clean cumulative Release candidate has SHA-256 `94784f164856c3fdedf743a2b459448b4d3fd3878b418e9d64db5fc98b6c8503`. Exact conflict-9450 profiling reduces the combined selection owner 24.73% and complete prefix cycles 0.442%. All 64 event rows are byte-identical to the frozen authority. Full Rank 27 is exact-state UNSAT in 373.696 seconds, and the mandatory B28+C5 gate passes 66/66 at 0.995586x predecessor PAR-2 with no invalidity, solve loss, or residue. Claude Code's authoritative fuzz report is PASS, completing the final promotion gate.

The BVE prerequisite is independently closed and pushed at commit `a07bc93`. Applying the isolated three-file repair over that clean baseline reproduces Release SHA-256 `94784f164856c3fdedf743a2b459448b4d3fd3878b418e9d64db5fc98b6c8503`, and focused policy/N1/trace/reset validation passes 8/8. Checked Rank-27 proof job `1647655` also passes with the accepted proof and proof-mode logical identity. Full Release is 73/74 with every functional, proof/model, trace, N1, boundary, cleanup, and reset test green; only the pre-existing opt-in documentation-size policy fails.

1 Source boundary and exact prefix

The focused selector starts at the cached VMTF search cursor and follows predecessor links until it finds an unassigned variable. The stable selector repeatedly removes assigned heap maxima until the root is unassigned. The new entry points keep those exact traversals, lazy pops, trace hooks, and queue cache updates, but replace the generic "no variable remains" backstop with a debug assertion of the caller-owned active invariant. The diagnostic no-extension arm still uses its existing total path. No decision phase, random-decision admission, heap/queue mutation, rule schedule, proof behavior, or strategy constant changes.

Matched profile `1647582` stops both binaries at conflict 9450. Their complete normalized non-time output has byte-identical SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`. This fixes every conflict, decision, propagation, rule tick/effect/order, schedule, learning, pruning, and retained-state aggregate. With zero lost samples, complete cycles fall 71.737B to 71.420B; combined queue/heap next-unassigned work falls 1.458B to 1.097B, and the focused queue owner falls 1.306B to 0.946B. Elapsed time falls 18.717 to 18.626 seconds.

Candidate-only trace 1647584 emits exactly 64 untruncated rows at SHA-256 7a6216bb86c23651 2dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1, byte-identical to the frozen common-state authority. It therefore directly preserves decision/rule/candidate order, exact tick and cost, scheduling, learning, pruning, and resulting state.

2 Long-horizon and mandatory gate

Node-18 Rank-27 job 1647585 returns exact-state UNSAT with accepted complete non-time SHA-256 1 4f7da2d94153a2afd1776ecab8e7e49e5c2c0613dc26c43714471b7ecbe6194. Internal time is 373.696 seconds and wall time 374.18 seconds. Against the immediately preceding exact BVE repeat at 375.797 seconds, it recovers 2.101 seconds entirely outside inprocessing; BVE is 15.522 versus 15.500 seconds and inprocessing 139.807 versus 139.800. The candidate is 1.985 seconds slower than the single best 371.711-second BVE observation, so one complete timing is supporting evidence, not an independent promotion claim.

Frozen full-width campaign 1647586/1647652 compares the exact BVE pin with this candidate over all Tier B plus the predeclared mechanism-relevant C5. All 66 tasks are valid. Each arm independently validates 17 SAT models and 16 UNSAT proofs against the original decompressed CNF. Every one of the 33 paired normalized non-time images matches. Candidate PAR-2 is 1592.114 seconds versus 1599.173, ratio 0.9955858; both arms peak at 2.917 GiB. There is no invalidity, solve loss, cleanup failure, or certificate residue. Summary SHA-256 is 9846d2c77f255628d495492d8a381e c788d7b3044d6fd13f217feb659b5e20f0.

Proof-producing job 1647655 uses the exact cumulative Release pin under two CPUs, 16 GiB, a 1,200-second solver limit, and a 4,800-second checker limit. It returns UNSAT in 522.462 seconds internal / 522.994 seconds wall. Pinned `drat-trim` verifies the 1,514,395,111-byte proof in 128.730 seconds with `s VERIFIED`. Proof SHA-256 is 0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e12744 2e50f47e25bea8bb2d; the complete normalized proof-mode image has accepted SHA-256 8cd3eda481 799871a179e1483a57413622bc7832c09432df705b2308bafc2da7. Both identities are byte-identical to the accepted BVE witness. Proof, input, checker output, and scratch are deleted with zero residue.

Claude Code exclusively prepares, submits, monitors, and classifies exact- source ASan/UBSan native-LRAT fuzz campaign 1647657. Its authoritative terminal report is PASS. Corpus replay is 27/27 with zero failures and zero semantic nonterminations. Random fuzz is 10,000/10,000 with zero failures: 5,667 SAT models are independently accepted against their original CNFs, 3,906 UNSAT native-LRAT proofs are accepted by pinned `lrat-check`, and Kissat agrees on all 9,573 verdict cases. The remaining 427 generated cases hit the declared five-second quick bound and carry no oracle claim; 25 capped diagnostic CNF bundles and the complete seed ledger are inputs, not failures or certificates. Required reach rows fire for Decide 1,280, SwitchMode 395, and Propagate 7,280 cases, directly exercising both focused and stable active pickers. Claude reports zero sanitizer finding, crash, invariant assertion, model/proof rejection, differential disagreement, certificate residue, or job scratch residue.

3 Disposition and next work

The candidate satisfies its exact-stage, exact-trace, full Rank-27, complete B28+C5, clean-source rebuild, focused local validation, and checked proof contracts. Promotion now requires only Claude Code's authoritative fuzz PASS and the final Release/report wrap-up; both are now complete. The public total selector remains the N1 authority; the optimized entry points belong only to DefaultPolicy sites whose active guard is explicit.

After this increment, the next source-backed broad owner is collection and rewatching. A conservative current-Kissat conflict-10000 profile spends 4.521B cycles in explicit collection/flush/rewatch operations, while the shorter exact Tactix prefix spends 7.451B. Existing Tactix disassembly assigns about 60% of `kernel_collect_garbage`'s local samples to a stable watch compaction followed by a second retained-entry walk that only forwards large Refs. A fused compact-and-forward primitive is the next bounded representation hypothesis, but no prototype starts before this increment is independently landed. After collection/rewatching, matched Substitute invocations are the next direct localization target, followed by the smaller residual Congruence difference and continuing broad search-quality convergence. Each repair is a separate exact-prefix, Rank-27/B+C, full checked gate, Claude fuzz, and report increment.